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SURVEY

An Exploratory Survey on the Use of Graph Algorithms in Analysis of Social Networks

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ABSTRACT A social network is a web of connections between individuals or groups. These expansive social networks can encompass thousands to millions of nodes, providing valuable insights into the structures and characteristics of these networks. *Social Network Analysis (SNA)* serves as the established methodology for exploring these structures, whether among individual entities, groups, or the entire social network. The primary objective of this analysis is to gain an understanding of the network's properties, whether considered as an integrated whole or through the examination of individual nodes. Typically, these networks are represented as graphs, where nodes symbolize the social entities and edges signify their relationships. Graph algorithms enable the study of these graphs and have become essential in social network analysis. Despite the extensive research into this area, there is a lack of a holistic view that consolidates the multitude of social analysis tasks and their associated graph algorithms into a single practical guide. This study identifies these tasks and provides a practical, step-by-step guide, supported by a case study that demonstrates the viability and usability of the framework, thereby bridging the theory-practice gap in social network analysis literature.

INDEX TERMS Social network analysis, graphs, graph algorithms, community detection, similarity, clustering, link prediction, social network analysis tasks.

I. INTRODUCTION

In the 2022 estimate of the global population by the United Nations [1], they reported that the world's population totals approximately 7.9 billion individuals. Within this demographic, an estimated 4.9 billion are active users of social media platforms [2]. The proliferation of these social media platforms has resulted in the abundance of relational data now available for study. Oliveira and Gama [3] posit that the relationships and interactions of an individual yield more insight than their standalone attributes. Studying these relationships is advantageous for a variety of applications, such as fraud detection, marketing, and so on. These interactions among entities viewed as a whole networked structure constitute social networks.

A social network is essentially a web of connections between individuals or groups [4]. It is formulated using relational data. Social networks are defined as a collection of

social entities encompassing individuals, groups, and organizations interconnected by relationships or interactions. In real life, one can obtain these relationship patterns from social media data, smartphones, proximity sensors, communication networks, private company databases, etc. Typically, these networks are represented as graphs, where nodes symbolize social entities and edges signify the connections forged among them [3], [5].

A conventional social network can encompass thousands to millions of nodes. *Social Network Analysis (SNA)* is the established methodology for exploring relationships between individual entities, groups, or the entire social network, providing valuable insights into the network structures and characteristics. The primary objective of this analysis is to understand the network's properties, whether considered as a whole or through the examination of individual nodes [4]. This approach allows for the exploration of the relationships among nodes and the implications of these relationships.

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Throughout this paper, we will use the term “*node*” to describe entities involved in social interactions, such as friends and followers [4]. The connections between the nodes are the “*edges*”. The architecture of social networks is the focal point of examination in Social Network Analysis (SNA) [3]. As an interdisciplinary methodology, SNA enables visualizing complex social structures and revealing otherwise undetected communication channels, often showcasing prospects for collaboration and identifying potential gaps in interactions that might have been challenging to perceive through other means [6], [7]. SNA involves a set of tasks critical to exploring and understanding network structures. Some of these tasks include, but are not limited to [3], [6], and [7]:

- Identification of influential entities within networks.
- Pinpointing *hubs* (highly connected nodes within a network) and *authorities* (nodes with significant influence, trustworthiness, or expertise) through often using link analysis algorithms.
- Discovery of *communities* (a group of nodes exhibiting stronger interactions with each other than with nodes outside the community) within networks.

There are two distinct methodological approaches to social network analysis [8], [9]:

- 1) *Content Analysis*. Content analysis involves activities such as gathering statistical data and interpreting textual information. Machine learning techniques are popularly applied in content analysis. Endeavors such as categorizing nodes in a network based on a particular theme are achieved using content analysis.
- 2) *Structural Analysis*. These methods are based on analyzing the graph structure topologies of social networks. When analyzing social networks, it is important to note that the structures formed by various elements are of equal significance to their respective categories. These relationships can be represented through graphs featuring diverse configurations of nodes and edges. Despite the limited set of node and edge types, their arrangements give rise to many unique structures, each with distinct connotations that can be deduced using graph algorithms.

This paper investigates the use of graph algorithms in SNA, with a focus on the structural analysis dimension. To achieve this goal, this study is organized as follows: Section II identifies the need for conducting a comprehensive survey on social networks and outlines the objectives of this work. Section III provides an overview of the key concepts within this survey and the historical role of graph algorithms in SNA. In Section IV, the survey methodology is assessed, while Sections V, VII, VI, and VIII include the analysis, discussions of findings, a representative case study, and our concluding statements.

II. MOTIVATIONAL RESEARCH QUESTIONS AND OBJECTIVES

A. PROBLEM STATEMENTS

1) RESEARCH GAP 1: COMPREHENSIVE UNDERSTANDING OF SOCIAL NETWORK ANALYSIS TASKS WITH GRAPH ALGORITHMS

While existing research has shed light on individual social analysis tasks, particularly the graph algorithms that underpin them, there is a void in the literature — a lack of a holistic view that combines the multitude of social analysis tasks and their associated graph algorithms. This gap calls for an examination that bridges theory and practical applications, expounding on how these algorithms can be harnessed synergistically to yield a more comprehensive understanding of social networks and their multifaceted applications.

To address this research gap, we pose the following research questions:

- 1) What are the critical SNA tasks that involve graph algorithms, and how do these tasks collectively contribute to our understanding of social networks?
- 2) Can a taxonomy or classification framework be developed to categorize social analysis tasks based on their underlying graph algorithms and the nature of insights they provide?
- 3) How do different graph algorithms interplay when applied to diverse social analysis tasks, and what synergistic effects can be observed regarding knowledge extraction from social networks?

2) RESEARCH GAP 2: BRIDGING THEORY AND PRACTICAL APPLICATIONS OF GRAPH ALGORITHMS

While extensive theoretical discussions have contributed to our understanding of graph algorithms, a notable research gap lies in translating these theories into practical, real-life applications. Many studies focus on the subtleties of graph algorithms in isolation, often overlooking the practical implications, including time and space complexities, when used in real-life applications. To address this research gap, we pose the following research questions:

- 1) What are the specific real-life applications of graph algorithms in SNA?
- 2) How do practical considerations, such as time and space complexities, impact the deployment of graph algorithms in real-world scenarios, and what are the trade-offs involved?
- 3) Are there best practices for people looking to utilize graph algorithms in practical applications within SNA, considering computational efficiency and resource constraints?

Through these research questions, we will fill in the research gaps and provide our perspective on how graph algorithms play a crucial role in improving our understanding of social networks via SNA.

B. RESEARCH OBJECTIVES

The primary objectives of this survey paper are as follows:

- 1) To Identify and categorize the key SNA tasks that exploit graph algorithms and explore how they contribute to an overarching understanding of social networks.
- 2) To develop a taxonomy or classification framework that categorizes social analysis tasks based on the graph algorithms they employ and the insights they provide, facilitating a structured view of the field.
- 3) To investigate the interplay between different graph algorithms when applied to diverse social analysis tasks, aiming to uncover synergistic effects that deepen our understanding of social networks.
- 4) To explore and document real-life applications of graph algorithms in SNA.
- 5) To assess the practical considerations, including time and space complexities, associated with deploying graph algorithms in real-world scenarios and highlight the trade-offs involved.
- 6) To distill best practices and guidelines that can aid practitioners in effectively using graph algorithms within SNA. How can they improve computational efficiency given resource constraints?

In order to accomplish these objectives, we carry out a survey of publications on graph algorithms in the context of SNA.

III. PRELIMINARY

A. KEY CONCEPTS

A graph (G) consists of two sets of finite elements V and E such that:

$$G = (V, E) \quad (1)$$

where V represents *vertices*, and E , *edges*. Each $e \in E$ is identified with a pair of *end vertices*, $e_i = \{v_i, v_j\}$ where $v_i, v_j \in V$ [10], [11]. In this case, v_i, v_j are *incident* with e_i , and v_i, v_j are described as *adjacent* vertices or *neighbors*. The *degree* of a node or vertex is the number of edges connected to it. A graph G_n is an n -vertex graph, i.e., $|V| = n$. Programmatically, graphs can be represented as an adjacency list or an adjacency matrix. Fig. 1 and Fig. 2 depict a typical adjacency list and matrix, respectively.

Algorithms are sets of statements, a procedure for solving *well-specified* computational problems. While the computational problem indicates the input and desired output, an algorithm defines the procedure for achieving the desired input/output relationship. A *correct* algorithm produces the intended output for every input [12]. Various real-life graph problems exist, just like with SNA. The graphical representations of these problems often exhibit a substantial vertex-edge ratio [10]. Navigating graph problems requires an understanding of graphical properties as developing efficient solutions typically involves analyzing a graph or testing for one of said graphical properties [10]. Thus, we must depend



FIGURE 1. Adjacency list.

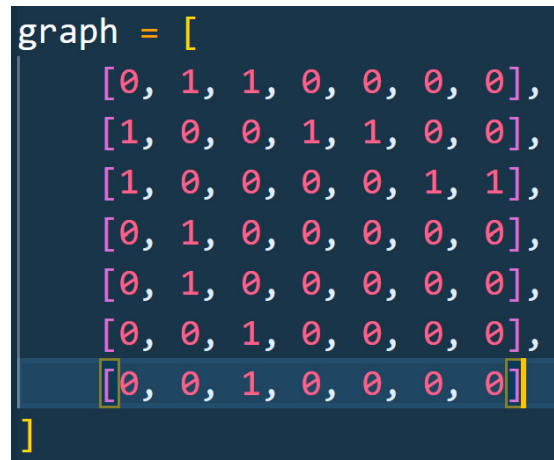


FIGURE 2. Adjacency matrix.

on *graph algorithms* to analyze these complex graphs. Graph algorithms leverage the principles of graph theory to analyze or manipulate graphs.

The primary considerations for every algorithm that determines its efficiency in solving a given problem are the *computational complexities* - *time and space*. Time complexity is a measure of the running time of the algorithm, and space complexity is a measure of the storage required for the algorithm during runtime. The complexity of a graph algorithm is a function of the size of the input. For a given graph (1), the size of the input is determined by two parameters: $|V|$ the number of vertices and $|E|$ the number of edges [10], [12]. Generally, to avoid ambiguity in graph algorithms within asymptotic notation, $|V|$ is represented by V and $|E|$ by E [12].

SNA has its origins in the Social Sciences. According to Freeman [13],

“the structural approach that is based on the study of interaction among social actors is called social network analysis” [13].

Progressively, researchers determined that these social actors, initially considered individuals, could also be groups

or organizations. Social networks are based on the concept that the patterns of social connections that the *actors* are involved in are of significance to them. Thus, SNA aims to identify and understand these patterns and explore the conditions that led to their formation and their impact on the actors [13]. Contemporary SNA is synthesized into the following characteristics [13]:

- It is motivated by the social ties between actors.
- It relies on real-world observations and measurements.
- It depends on graphs to represent the relationships.
- It relies on mathematical and computational models to analyze and interpret data.

B. HISTORICAL OVERVIEW OF THE ROLE OF GRAPHS AND GRAPH ALGORITHMS IN SOCIAL NETWORK ANALYSIS

1) PREHISTORY - THE PERIOD ENDING IN THE 1920s

The use of graph-like depictions in studies that examine the relationships between the objects of study (*e.g.*, SNA) dates back to as early as the ninth century [13] in a study where the researchers used tree-based images to depict kinship pictorially [14]. In the late 19th century, Alexander Macfarlane constructed visual representations of different degrees of kinship [13], formalizing the British model of the kinship system. These studies used points to represent the actors and lines for the relationships between the actors [13]. In his research on interacting directorates, John Atkinson Hobson (1894/1954) also produced hypergraphs to visualize his data [13]. The mathematical models these researchers used were mainly algebraic and probabilistic [13].

2) THE 1930s - SOCIOMETRY

This era featured *sociograms*, a visual model for representing social networks using points and lines [15]. Jacob Moreno (psychiatrist) and Helen Jennings (psychologist) used these sociograms [13], [15], [16] in their experimental inquiry into the evolution and organization of groups and the individuals within them. This approach to SNA was named *psychological geography or sociometry* [15]. Their research is the precursor for modern SNA. Once again, the mathematical/computational models used in sociometry were just probabilistic (*Paul Lazarsfeld model* [13]).

3) 1940 TO 1960s - THE BIRTH OF SOCIAL NETWORK ANALYSIS

There was greater interest in social structural research during this period, particularly among educational institutions. Various research efforts were undertaken at Harvard University, and graphic images were employed in their reports [13]. Hypergraphs showing *clique* membership within organizations, groups, and directed graphs indicating the flow of social connections were highlighted [13]. The Bavelas Group at MIT developed a formal model for *centrality* and used visual expressions derived from graph theory to represent the social patterns they discovered [13]. At the University of Michigan, Frank Harary focused on the applications

of graph theory to the problems of social structure [13]. From this point on, in subsequent research, the graphic theoretical model was utilized in social network research and analysis; more than just the graphic images as a visual representation, their properties became intertwined with said research conclusions.

4) 1970S - EARLY 2000s

Prior to this point, research into social networks was a scattered effort [13]. However, with the advent of technology, closer collaboration in the field became possible. Moreover, with the advent of computer programs, the field rapidly developed. These computer programs employed some basic graph algorithms and mathematical models to automate the manual analysis of social networks. The earliest recorded instance was the computer program developed by James S. Coleman and Duncan MacRae in the late 1950s, later extended by Seymour Spilerman in 1966 [13]. The researchers designed the program to find closely linked individuals in a social network (*now community detection/clustering*). In 1971, BLOCKER, developed by Gregory Heil and Harrison White, emerged to find individuals occupying the same positions in a social network (*distributionality of a graph*) [13]. As the field matured, multipurpose SNA programs such as UCINET and SONIS were introduced [13].

5) 2000s TO PRESENT

The rise of social media and improved computing power have driven a paradigm shift in the analysis of social networks. Larger chunks of data are now readily available. Consequently, researchers have moved away from building algebraic and graph-theoretic models to complete dependence on computational procedures [16]. More efficient graph algorithms are being discovered and employed, making the field truly interdisciplinary.

This research paper investigates the current use of graph algorithms and their specific applications in SNA tasks.

C. SOCIAL NETWORK ANALYSIS TASKS

SNA entails various activities depending on the type of social network under consideration, the application area, and the purpose for analysis. However, a few fundamental tasks are common across most network structures and analytical intentions. Higher-level SNA tasks often depend on some combination of the algorithms and models from these fundamental tasks. The fundamental tasks covered in this survey are not exhaustive but are drawn from the publications included in the survey. They are:

- 1) Graph Sampling
- 2) Link Prediction
- 3) Centrality
- 4) Community Detection & Clustering
- 5) Similarity
- 6) Graph Visualization

IV. SURVEY METHODOLOGY

This survey critically analyzes existing literature on graph algorithms for SNA. The population for this survey is all existing publications on graph algorithms within the context of SNA. We selected a sample from the population according to our selection criteria.

A. CRITERIA FOR SELECTING LITERATURE

In choosing our data sources for the survey, we considered the following criteria:

- Relevance: The publications were selected based on their relevance to our objectives.
- Publication Source: Preference was given to publications from peer-reviewed journals, books, and reputable conferences to ensure the quality of the publication.
- Recency: We prioritized more recent publications to capture the latest developments in the field.
- Diversity: We included a diverse set of social analysis tasks in our sample size to provide a comprehensive view of the field.

B. SEARCH STRATEGY

We used IEEEExplore, ACM Digital Library, ScienceDirect, and Google Scholar to retrieve relevant literature. Within these databases, we used keywords relevant to our objectives, including *graph algorithms* to *centrality*, among others. We restricted our sample to papers published in the last 20 years to ensure that we covered the most recent advances in the field. We selected a total of 43 papers that fit our search criteria.

C. REVIEW AND ANALYSIS METHODOLOGY

We implemented our screening process in two phases. First, we reviewed the title and abstract of the paper to eliminate irrelevant publications. Subsequently, we reviewed the pertinent papers in accordance with our selection criteria. Data from our selected literature were extracted. In particular, we extracted the time and space complexities of the mentioned graph algorithms and the SNA task directed towards them. We employed a thematic synthesis approach, and we were able to find the patterns across our literature. These findings have informed our survey.

V. DISCUSSION

A. GRAPH SAMPLING

Graph Sampling enables the analysis of large-scale networks by developing representative subsets of the more extensive network. The rise of the internet and social media has allowed for vast social networks, which by default are complex and overwhelming for direct analysis. Consequently, graph sampling algorithms are used to *crawl* the extensive networks and generate *snapshots*, *subgraphs*, or *motifs* that accurately capture the essential characteristics of the larger network. These subsets are used instead of the original extensive network for further analysis.

Unlike other graph algorithms, time and space complexities are not the most important metrics for evaluating graph sampling algorithms. Ye et al. [17] stated some factors that are often considered more important in assessing the efficiency of graph algorithms as regards their functionality:

- 1) Bias: Do the subgraphs' statistical properties effectively represent the original graph's characteristics? [17]
- 2) Efficiency: How quickly do we discover our nodes and links? [17]
- 3) Sensitivity: How does it react to different social network structures and missing links or *black holes*? [17]

Based on our survey, another factor is how accurately the algorithm can preserve the original graph's structural properties in the subgraphs.

As identified in our survey, some modern algorithms for this task are summarized in Table 1.

- **Metropolis-Hastings Random Walk (MHRW) with a Modified Sampling Procedure:** Bipartite graphs are used to represent two-node type networks. For example, a social network representation of eBay (see Figure 3) that models sellers and buyers is a two-node type network, where the node types are *buyers* and *sellers*.

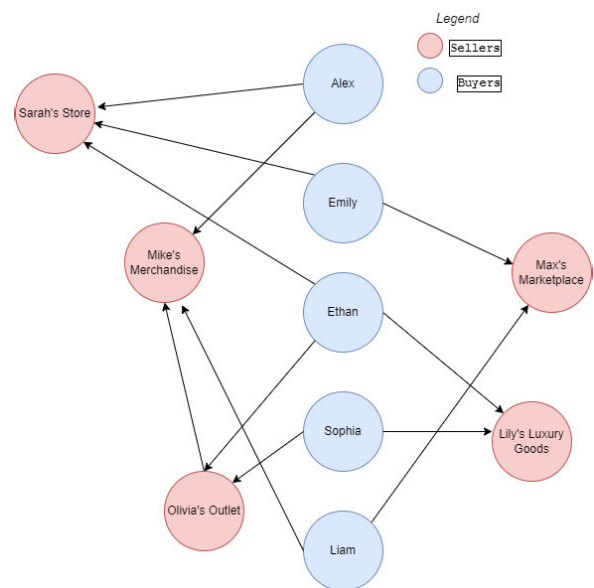


FIGURE 3. Sample Two-Node Network of Sellers and Buyers represented using graph coloring - Bipartite Graph.

The MHRW, with a modified sampling procedure, extends the legacy MHRW algorithm for sampling bipartite graphs, tweaking it to improve efficiency. Specifically, this algorithm features an extension to the sampling procedure of MHRW such that [18]:

- 1) Step 1: Select a random node V from the graph as the initial node.
- 2) Step 2: Generate a random number $P \in (0, 1)$

TABLE 1. Graph sampling algorithms.

Algorithm Name	Operational Aspects of the Algorithm	Complexity
Metropolis-Hastings Random Walk (MHRW) with a modified sampling procedure [18]	- Specific to bipartite (two-mode) social networks. - Reduces bias towards nodes with more edges. - Modified sampling procedure improves performance over regular MHRW for two-mode networks.	- Time: N/A - Space: N/A
Respondent Driven Sampling (RDS) or Re-weighted Random Walk (RWRW) [19], [17]	- Demonstrated efficiency in network structures where connections are not strictly organized, e.g., overlay networks. - Efficient in initial sampling. - Performs better than MRW in network structures where the distribution of edges per node is highly skewed, and there are clusters. - Bias introduced by random walks is corrected by appropriate re-weighting.	- Time: N/A - Space: N/A
Metropolis-Hastings Random Walk (MHRW) [20]	- Uniform sampling such that each node in the graph has an equal probability of being visited. - Produces versatile datasets.	- Time: N/A - Space: $O(1)$
Breadth-First Search [20], [17], [21]	Known to introduce bias towards high-degree nodes (<i>Data Structures: 1) Adjacency List, 2) Adjacency Matrix</i>)	- Time: $O(V + E)$ - Space: $O(V^2)$
Snowball Sampling [17], [21], [22]	- Efficient for complex network structures - Complexity is specific to the process and network. It would be based on the number of iterations and the final sample size.	- Time: N/A - Space: N/A
Random Walk (RW) [17], [20]	- Leads to a significant level of bias. - Allows for node revisiting. - Complexity is specific to the process and network. It would be based on the number of iterations	- Time: N/A - Space: N/A
Social Graph restoration via Random Walk (RW) Sampling [23]	- Enables prediction of the visual representation of original graphs from subgraphs obtained by a random walk. - Uses dK-series. - Utilizes the subgraphs along with the local structural properties.	- Time: N/A - Space: N/A

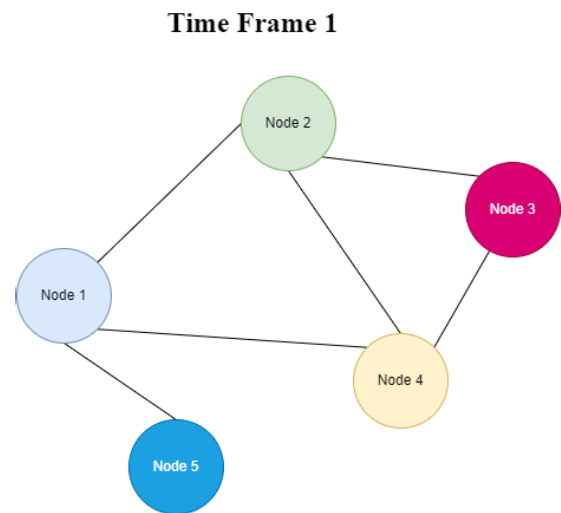
- 3) Step 3: For each neighbor node W of the current node V , a cumulative probability q is calculated based on the the previous node's q value and the degree of W . Compare P with q for each neighbor node. If $P \leq q_W$, then the current node becomes W , otherwise, the algorithm repeats the process with other nodes. The sampling procedure terminates when the stopping criterion is met.

$$|E| \times \text{SamplingRatio} \quad (2)$$

The originators determined that a sampling ratio above 0.8% yielded better performance [18].

This algorithm is best suited to sampling bipartite graphs.

- **Respondent Driven Sampling (RDS):** Dynamic unstructured P2P overlay networks are characterized by dynamic connectivity between the nodes, with each peer/node able to leave the network at will [19]. Figure 5 and Figure 4 depict an unstructured P2P network at two instances in time. In time frame 1, all nodes are connected, but in time frame 2, nodes 2 and 3 have left the network, and node 5 has established connections with Nodes 1 and 4. As seen from Figure 4, network graphs of these types are decentralized by nature. Hence, representative samples may only be obtained by crawling local neighbor interactions to determine a full

**FIGURE 4.** Sample P2P Unstructured Network at Instance 1 in time.

snapshot per time. A social network of this type could be obtained from online gaming platforms, e.g., Call of Duty. A game player could decide to host multiplayer games in which players can leave at will, in no particular order, and without a central node/player. Sampling such networks for specific peer properties without exhaustively visiting all nodes can be accurately

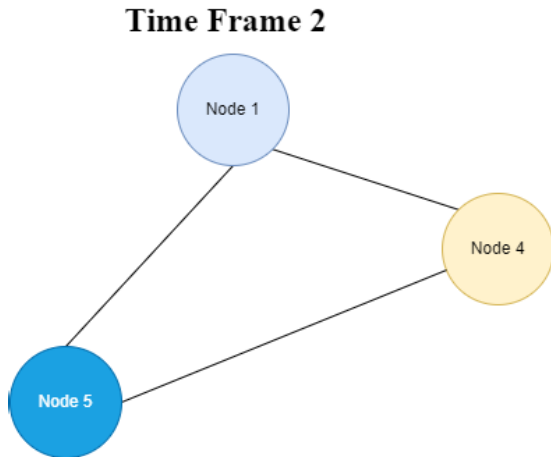


FIGURE 5. Sample P2P unstructured network at instance 2 in time.

done using Metropolis-Hastings Random Walk, but the progenitors found that RDS performs better in settings with significant variation in node connectivity and clustering [19]. An accurate snapshot, however, is a first step and an efficiency determinant in the practical use of RDS [19] for sampling unstructured P2P networks. Respondent Driven Sampling (RDS) is a derivative of the Snowball Sampling [19]. The methodology of the algorithm is summarized as follows [19]:

Say we want to estimate the proportion of nodes with a specific property in a network; for example, we want to determine the number of players in an online gaming network who are Italian:

- 1) Step 1: The nodes are partitioned into groups.
- 2) Step 2: RDS implements random walk, so nodes are visited and could be revisited over multiple steps, such that the properties of the visited nodes are recorded in each step.
- 3) Step 3: Estimate the proportion of Italian nodes in each group using the Hansen-Hurwitz estimator to account for biases in the sampling process.

The RDS estimator improves in accuracy with each visited node, converging to the true proportion of nodes (Italian) per group as more nodes are visited. The likelihood of visiting certain nodes is determined by their importance (weight) in the network structure.

- **Metropolis-Hastings Random Walk (MHRW):** The Metropolis-Hastings Random Walk algorithm is a legacy algorithm found in practice to perform well in obtaining uniform distribution representative samples of large social networks (e.g., Facebook) [20]. Unlike traditional algorithms such as the Breadth First Search (BFS) and the Random Walk, which cause significant bias within the samples towards the higher degree nodes, MHRW samples a given network in such a way that approximates a uniform distribution (each node has an equal chance of being selected despite characteristics or weight considerations) of the network making it

especially useful for online social networks known to have millions to billions of nodes.

To achieve a uniform distribution, the algorithm adjusts the transition probability between nodes based on the degree of the node and some minimum value [20]. For nodes y , V and W , where k_v is the degree of the node V the Metropolis-Hastings transition probability P^{MH} is [20]:

$$P_{v,w}^{MH} = \begin{cases} \frac{1}{k_v} \cdot \min\left(1, \frac{k_v}{k_w}\right) & \text{if } w \text{ is a neighbor of } v, \\ 1 - \sum_{y \neq v} P_{v,y}^{MH} & \text{if } w = v, \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

Given the described characteristics, this algorithm is well-suited for sampling very large social networks to obtain the most accurate representative sample.

- **Breadth-First Search (BFS):** The Breadth-First Search algorithm is a legacy graph traversal algorithm. Researchers have widely studied its applicability for crawling online social networks. Due to its traversal approach, BFS is prone to bias toward higher-degree nodes. As the name states, it explores all the neighbors of a given node (breadth) before proceeding to neighbors of neighbors.

This weakness implies that when applied to a typical social network, BFS cannot obtain an accurate representative sample. Rather, the sample will reflect nodes with a higher number of connected nodes (i.e., for example, in an online social network, individuals with a greater number of followers). In practice, the BFS algorithm would be ideal for influence propagation tasks or identifying the most influential nodes in a network. It could also be effective for sampling social networks where every node has equal weight (e.g., a simple communication network where every user has equal rights to communication). However, these uniformly distributed social networks with equally weighted nodes are rarely obtainable in real life.

- **Snowball Sampling:** Snowball Sampling is popular in social sciences research for selecting research participants based on some criteria. In the snowball sampling participant selection procedure, the researchers identify an initial set of participants and then request additional participant suggestions from the initial group, a chain-referral process [22]. In social network sampling, Snowball Sampling involves selecting a seed node and performing a Breadth-First Search [17] to achieve the desired sampling property. The final sampled network reflects the links between the selected nodes. Applying the Snowball Sampling algorithm can be considered in scenarios where the required sampling ratio is small, as it is more likely to yield an accurate representative sample with a smaller sampling ratio [17] compared to other

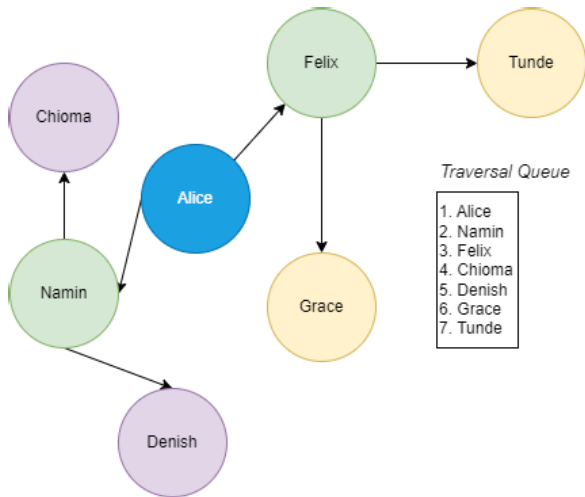


FIGURE 6. BFS Traversal.

algorithms. The algorithm is also adept at identifying hubs within a network (nodes with several neighbors), most likely due to its use of BFS algorithm, making it ideal for opinion leader identification, influence propagation, and similar or derivative tasks. However, the snowball sampling algorithm tends to oversample hubs [17]. This weakness must be taken into account when applying the algorithm in practice. Ye et al. [17] opined that Snowball Sampling is ideal for web sampling.

- **Random Walk (RW):** Similar to Breadth First Search (BFS), random walk sampling is a legacy sampling algorithm. As its name suggests, the process involves visiting random nodes with the possibility of revisiting each node. Random walks are a finite Markovian process in which both the forward and backward passes (future and past) behave similarly; this reflects the steady-state behavior of Markov chains. Random walks have been extensively studied in graph theory. However, they exhibit a calculable bias toward higher-degree nodes. Hence, researchers have extended and modified the basic random walk sampling algorithm to create improved algorithms such as the Re-weighted Random Walk (RWRW, Respondent Driven Sampling) and MHRW. Gjoka et al. [20] confirm that random walks have been evaluated in their applicability for web sampling, P2P sampling, and online social network sampling.
- **Social Graph Restoration via Random Walk (RW) Sampling:** This algorithm attempts to restore an original graph from a subgraph, mimicking the key local and global structural properties of the original graph as closely as possible [23] while retaining the structure of the subgraph. Summarily, the algorithm achieves this by adding nodes and edges to a subgraph sampled by a random walk¹ and using five (5) local structural

¹known for its bias towards higher degree nodes.

properties: the number of nodes, average degree of nodes, probability of distribution of degrees, probability of distribution of degrees of pairs of connected nodes, and degree dependent clustering coefficient which is the measure of how the probability that the of the neighbors of a given node are connected, varies with the changes in the degree of a node [23]. Compared to existing graph sampling algorithms such as BFS and Snowball Sampling, this proposed algorithm was shown to be more efficient at preserving local structural properties like the number of nodes, average degree of nodes, and neighbor connectivity [23]. Hence, in instances where these specific structural properties are desired to be maximized, such as influence propagation and recommendation systems, this algorithm can be used to obtain quick and accurate results. Furthermore, the algorithm displayed better accuracy for global structural properties: average shortest-path length, shortest-path length distribution, diameter, degree-dependent betweenness centrality, and the largest eigenvalue of the adjacency matrix. However, this algorithm was largely inaccurate in preserving structural properties like the network clustering coefficient and the degree-dependent clustering coefficient [23].

B. LINK PREDICTION

Link prediction in SNA involves estimating the probability of association between nodes in a social network. According to Huang and Ma [24], these links may either be current but undiscovered links within the network or links with a chance of being created between nodes in the future. The existing characteristics of the network structure are analyzed along with some prediction algorithms to make these predictions [24]. Link prediction algorithms help monitor interactions between nodes, friend suggestions, and product suggestions [25]. Big companies like Facebook and Amazon use these techniques to enhance user experience and increase their revenue [25].

The algorithms used in this task, as identified from our survey, are summarized in Table 2.

- **Adamic-Adar(AA)** The Adamic Adar Algorithm predicts links between nodes by combining the set of common neighbors and determining their weights. The similarity value (Adamic Adar Index) between two nodes, M and N , is computed using the formula [26]:

$$AA(M, N) = \sum_{p \in N(M) \cap N(N)} \frac{1}{\log |N(p)|} \quad (4)$$

where:

- $N(N)$ is the set of neighbors of node N .
- $N(M)$ is the set of neighbors of node M .
- $|N(p)|$ is the degree of the common neighbour p .
- $p \in N(M) \cap N(N)$ represents the common neighbours between M and N .

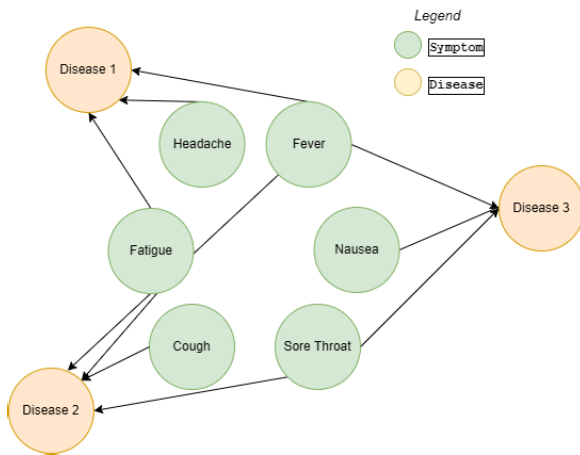


FIGURE 7. Adamic Adar link prediction.

Hence, if nodes M and N share common neighbors of low degrees, they are more likely to form a link. Given the nature of this algorithm, we can see that it would be suitable for analyzing social graphs with several small clusters to predict possible links. The example shown in Figure 7 indicates several closely related symptoms that could easily be misdiagnosed as Diseases 1, 2, or 3. Assuming a patient presents some symptoms, the AA algorithm could be used to determine the most likely disease.

- **Common Neighbour (CN):** As the name states, this metric predicts links between nodes based on the nodes they directly connect with, i.e., common neighbors. The ideology is that the more the neighbors two nodes A and B share in common, the more likely A and B can be linked to each other in the future [25], [27]. It is the simplest of the link prediction approaches. However, one can question its accuracy given that it assumes all nodes have equal weights, which is unlikely in most social network contexts. Despite this, it could be applied to SNA activities in which some general and quick link prediction assumptions need to be made without a focus on accuracy. Using the CN metric for link prediction is applicable to small, dense networks or homogenized networks where all nodes actually have equal weights, for example, a mesh network of routers, in which each router is connected to an equal number of routers.
- **Feature Integrated Node Neighbour Approach (FINN):** The Feature Integrated Node Neighbor approach attempts to account for link prediction based on common neighbors and features of the nodes by using a combination of the Jaccard Coefficient and Adamic Adar to formulate a similarity metric [27]. To avoid false positives, FINN uses a threshold similarity value above which there is a definite probability of a link between two (2) nodes under examination [27]. Using FINN mitigates the constraints that occur when using just the common neighbor approach to link prediction, which

ignores the potential weights of the connected nodes, or when using the profile feature approach, which can lead to insufficient data or many false positives. Link prediction using FINN is suitable for social networks with diverse node features, which is usually the case with social networks representing people and their connections, such as Dating Applications. Such networks typically benefit from a combined consideration of the structural connections and individual attributes, leading to more accuracy in predicting links between nodes.

- **Jaccard Coefficient:** For two nodes M and N , the Jaccard Coefficient is a similarity value that takes into account the common neighbors between M and N :

$$J(M, N) = \frac{|N(M) \cap N(N)|}{|N(M) \cup N(N)|} \quad (5)$$

where:

- $N(M)$ is the set of neighbors of node M .
- $N(N)$ is the set of neighbors of node N .
- $|N(M) \cap N(N)|$ is the intersection of common neighbors between nodes M and N .
- $|N(M) \cup N(N)|$ is the union of unique neighbors of both nodes M and N .

The higher the Jaccard Coefficient value, the more likely the nodes under consideration can be linked. Computing the Jaccard Coefficient is computationally expensive for larger networks because it means calculating intersections for all pairs of nodes in the network. Therefore, the similarity measure is not ideal for much larger social networks [28]. However, some algorithms have been proposed for improving the computational efficiency of the Jaccard Coefficient algorithm, such as the parallel algorithm proposed by Hassani et al. [28].

C. CENTRALITY

Centrality is a set of metrics commonly used within SNA. It is not a task, but the algorithms used to determine centrality are essential to several SNA tasks, such as recognition of opinion leaders, influence maximization, structural hole analysis, community detection, etc. The centrality metric is a quantitative measure of the importance of a node in a social network [30], [31]. The composite centrality metrics include [30]:

- 1) Degree Centrality: Ranks nodes based on the number of connections. The more the connections, the greater the degree of centrality.
- 2) Eigenvector Centrality: This metric estimates the structural importance of a node based on its connection to other *important* nodes.
- 3) Page Rank Centrality: This metric assesses the quality and quantity of the nodes and assigns scores accordingly.
- 4) Betweenness Centrality: Calculates the importance of a node based on the number of shortest paths that cross through it. Such nodes are central connectors in the network.

TABLE 2. Link prediction algorithms.

Algorithm Name	Operational Aspects of the Algorithm	Complexity
Adamic-Adar (AA) Algorithm [24], [26]	Based on node similarity using common neighbors.	Time: $O\left(\sum_{i=1}^{ V } \text{degree}(V_i)\right)$ Space: $O(V + E)$
Common Neighbor (CN) [25], [26]	- Most popularly used - Simple - Based on the idea that more shared neighbors between 2 nodes make a connection more likely. (Data Structures: 1) Adjacency List, 2) Adjacency Matrix)	Time: 1) $O(V^3)$ Space: 2) $O(mn)$ with m rows (V) and n columns.
Feature Integrated Node Neighbor Approach (FINN) [27]	- Uses a combination of the Jaccard coefficient and Adamic Adar for prediction based on node connections and user-specified features. - Well-suited to datasets where profile features and connectivity information are available.	Time: $O(V) + O(E) + O(d + f)$ where, d = number of neighbors, f = number of features Space: $S(V, E, E', F) = O(V + E + V \cdot E' + V \cdot F)$ where, E' = Size of Ego network, F = average size of Feature Set
Jaccard Coefficient Algorithm [25]	Based on the idea that a pair of nodes with shared neighbors is more likely to form links.	Time: $O(\max(\text{degree}(V_i), \text{degree}(V_j)))$ Space: $O(V + E + \max_degree)$
Preferential Attachment (PA) Algorithm [25]	- Based on the idea that nodes with higher degrees are more likely to accept more connections. - Has the lowest computational complexity.	Time: Worst Case - $O(V^2 \times d)$ where, d = average number of connections per node, Space: N/A
Resource Allocation (RA) Algorithm [25], [20]	- Similar to the AA algorithm. - Discounts high-degree common neighbors more than the AA algorithm. (Data Structures: 1) Adjacency List, 2) Adjacency Matrix)	Time: $O(V^3)$ Space: $O(mn)$ with m rows (V) and n columns.
CNGF Algorithm [29]	- Based on node guidance capability, the density of the common neighbor subgraph. - If a common neighbor occupies a greater proportion in the neighbor set of a node, it implies that this common neighbor has a higher ability to generate new links between the node and other nodes.	Time: $O(N^2)$ where N is the maximum of all the nodes' degrees. Space: N/A
KatzGF Algorithm [29]	- More paths between nodes in the Katz algorithm mean a higher chance of a new link between them. - Can distinguish the contribution of different nodes effectively than the Katz Algorithm. - Not suitable for extensive networks.	Time: $O(V^n)$ Space: N/A

- 5) Closeness Centrality: This metric evaluates nodes based on the information flow. Nodes with a higher closeness score have a greater proximity to others.
- 6) K-shells Centrality: Involves organizing networks into layers based on the degree of the nodes.
- 7) Group Centrality: Involves centrality metrics applied to groups of nodes.

The algorithms, identified from our survey, are summarized in Table 3.

D. COMMUNITY DETECTION/CLUSTERING

The topic of community detection is vital in SNA. A community within a social network is a collection of strongly interconnected nodes that are sparsely connected to other nodes in the network [36]. These community structures are also referred to as *clusters* [37]. Community detection enables more efficient analysis of large social networks. It also allows for topological structure recognition among other high-level SNA tasks. The more general concept of clustering goes

TABLE 3. Centrality Algorithms.

Algorithm Name	Operational Aspects of the Algorithm	Complexity
Energy-Based-Centrality-Measure (EBCM) [32]	- Based on the graph energy of a network, a measure of the absolute sum of eigenvalues of a graph. - Measures centrality based on sub-graph connectivity.	Time: $O(n^{3.373})$ where n is number of nodes. Space: N/A
Crawling-based algorithm for estimating the top-k nodes with the highest betweenness centrality in online social networks. [32] Brandes Algorithm	- Accurate when the sample size is minimal. - Crawling-based algorithm to estimate top-k-nodes with high betweenness centrality. - Calculates betweenness centrality. - Fast Algorithm - Not practical for vast networks	Time: N/A Space: N/A
Page Rank Algorithm [33]	- Each node is given a page rank value - The first K-nodes, when the PR values are arranged in order, are the top-K nodes.	Time: $O(VE)$ for weighted graphs $O(VE + V^2 \log V)$ for unweighted graphs Space: $O(V + E)$
Degree Centrality [33], [34]	- Employs heuristic strategies to determine centrality. - Sum of all first-order neighbors of a node. - DC has a significantly lower propagation range than the greedy algorithm for large networks.	Time: $O(V \log V)$ Space: N/A
Greedy Algorithm [33]	- Maximizes the spread of cascades. - Makes local optimal choices	Time: $O(K^2 V^2)$ where K is the spread. Space: N/A
Closeness Centrality [34]	- Utilizes BFS - Applied Sequentially	Time: $O(V^3)$ for unweighted graphs Space: N/A
Centrality Using Degree and Paths Algorithm [35]	- Balances the advantage of degree centrality and number of paths of a given node - Performs optimally when using path lengths of two for most networks.	Time: $O(V^2)$ Space: N/A

beyond SNA, but the application of its algorithms towards community detection is considered in our survey. There are different methods of community detection, as highlighted by Rajita and Panda [38]. As identified from our survey, Table 4 summarizes the algorithms used for this task, which span all the methods of community detection.

For more algorithms, several surveys have been carried out on community detection. Rajita and Panda and Varsha and Patil [38], [39] provide an overview of the field, listing more algorithms than have been listed in this paper.

E. SIMILARITY

In structural SNA, similarity determines the likelihood between nodes, subgraphs, or even entire social network graph topologies [8]. Similarity graph algorithms are multi-task and can be applied to various SNA tasks requiring some degree of comparison. Most of the link prediction algorithms stated in this paper are also similarity algorithms, as the probability of a pair of nodes being linked in the future can sometimes be based on their similarity in terms of

topology and other nodes they are connected to, among other factors. Though similarity graph algorithms have applications beyond SNA, this paper is focused on their use in this field. In Table 5, we state a few more of the similarity graph algorithms we encountered in our survey. Foundational to similarity-based network analysis, especially in terms of data retrieval (e.g., you want to query a social network for the closest node or subgraph to a given node/subgraph), there are concepts such as proximity graphs, similarity queries, feature extraction, and the search algorithms for these proximity graphs [45]. Shimomura et al. [45] provide a comprehensive survey on the time complexities of algorithms used for constructing proximity graphs and search algorithms.

F. GRAPH VISUALIZATION

Visual representations of social network graphs are one of the premier steps towards analysis. Graph visualization is a multi-use, multifaceted area that has drawn much research interest. Graph algorithms are utilized in graph visualization. They enable an accurate representation of

TABLE 4. Community detection algorithms.

Algorithm Name	Operational Aspects of the Algorithm	Complexity
Label Propagation Algorithm [39]	- Not stable - No unique solutions - Uses local structure for community detection.	Time: $O(m)$ Space: N/A
LPA-Based Algorithm (LPA) [37]	- Improves the accuracy and stability of LPA. - Utilizes the tie-break strategy for initial node selection.	Time: $O(E)$ Space: N/A
P-Metric Algorithm [36]	- Detects communities using the P-metric P is based on common neighbors. P-metric demonstrated better performance than other metrics used in local community detection algorithms.	Time: $O(V^2 d^3)$ where d is the average degree of nodes inside the community. Space: N/A
Louvain Algorithm [40], [36]	- Fast, efficient, robust. - Good for large static networks. - Forms communities starting with an initial node, then others based on the modularity.	Time: $O(V \log V)$ Space: N/A
Clique-Louvain Algorithm [40]	- Solves the excessive time complexity problem of the Louvain algorithm - Time complexity is better than Louvain on synthetic datasets - Time complexity better than Louvain on very large datasets	Time: Linear Time Space: N/A
Enhanced Girvan-Newman Algorithm [41]	- Enables multiple edge removal - Reduces the number of iterations for each edge removal	Time: Worst Case - $O(V^2 E)$ Space: N/A
Incremental Dynamic Community Detection Algorithm based on Node Participation (NPICD) [42]	- Useful in dynamic social networks - Based on node participation and node adaptability to the community. - Utilizes the Louvain algorithm for the clustering step within the algorithm (Legend: 1) Initial Community Detection 2) Network Slicing in the future 3) Incremental Clustering for Network Changes 4) Edge Deletion 5) Edge Addition 6) Optimizing Small Communities)	Time: $O(N \log N)$ $O(MN)$ $O(n \log n)$ $O(m^2)$ $O(m^2)$ $O(C_1 \times C_2)$ where, M = number of edges in the network, N = number of nodes, C_1 = number of small communities, C_2 = number of connected communities, n = size of smaller communities, m = the number of edges incident to a node. Space: N/A
Local Spectral Algorithm (LOSP) [43]	- Local spectral subspace-based approach for community detection. - Seeks a sparse indicator for the target community through l_1 norm minimization over the Krylov subspace.	Time: N/A Space: N/A
Local Community Detection based on NGC Nodes (LCDNN) [44]	- The NGC node for each node is the closest vertex with the most centrality. - Could be expanded for multiscale local community detection.	Time: $O\left(\sum_{i=1}^n d \cdot \left[d^L + \log(i \cdot d)\right]\right) =$ $O\left(nd \cdot \left(d^L + \log(nd)\right)\right)$ where, L = number of hops from node to NGC node. $v = i$ th node added to local community d = degree of nodes in the community. Space: N/A

networks of all sizes, typically adopting some form of illustration or iconography to depict the nodes and edges in a network while preserving features. However, several tools utilize these algorithms to represent graphs. These tools span various graph-related fields and represent graphs based on user requirements. Common tools include Pajek [50], NetworkX [51], NetworKit [52], Gephi [53], Graphviz [54], Netlytic [55], Allegrograph [56], GRATIS [57]. It is important to note that most of these tools extend beyond graph visualization to enable analysis, incorporating other graph algorithms from other tasks for added functionality. Practitioners often use these tools to perform analyses on social networks. Faysal and Arifuzzaman [58] perform a comparative analysis of some of these tools.

VI. ANALYSIS AND SYNTHESIS

The fundamental social analysis tasks that we have identified through this survey regarding their use of graph algorithms have been thoroughly discussed in V. Graphs, graph theory, and graph algorithms of varying complexities and purposes are well-established. Due to the nature of the SNA field, we have found that these graph algorithms are not strictly tailored to the task for which they were identified. Some of them are used across the fundamental tasks we have listed, and some combinations of these algorithms could be applied to higher-level tasks. Some of these higher-level tasks include:

- Topological Structure Analysis
- Influence Propagation
- Opinion Leaders Recognition

TABLE 5. Similarity algorithms.

Algorithm Name	Operational Aspects of the Algorithm	Complexity
NetSimile [46]	- Based on the structural features of the nodes as well as social theories. - Generates a single signature vector for each group. As a measure for comparison.	Time: N/A Space: N/A
Model of Models (MOMO) [47]	- A similarity framework that follows the minimum description length principle, capturing the similarity between input graphs in terms of a common model. - Splits the problem into 2 parts and uses BEPPO and GIGI algorithms for the parts. - BEPPO performs graph summarization. Minimizes the graph to the smallest representative topology. - GIGI develops a common model based on the graph summaries from BEPPO. - If the models do not align, a MAXIMALGREEDY algorithm is used to develop a node overlap graph. - Depending on the number of edges, performance could be linear.	Time: BEPPO $O(VE)$ GIGI - $O(1)$ Hence MOMO - $O(VE)$ Space: N/A
Neighbor Matching Granli Similarity [48], [49]	- Node similarity algorithm. - Outputs a node similarity matrix. - Evaluated based on accuracy. - Utilizes the concept of Neighborhood matching	Time: N/A Space: N/A
Zager-Veghese Graph Similarity Algorithm [48]	- Nodes are considered similar if the neighbors are the same. - Utilizes the concept of Neighborhood matching - It is an iterative process until convergence, where similarity has been established. It is an iterative process until convergence, where similarity has been established.	Time: N/A Space: N/A

TABLE 6. Graph visualization algorithms.

Algorithm Name	Operational Aspects of the Algorithm	Complexity
Hierarchical Incremental Mode Analysis for Graphs (HIMAG) [59]	- Detects subgraphs across various resolutions, disregarding sparse regions. - Outputs an HMA matrix [60] - Authors developed a corresponding tool to work with the algorithm Gene DIVER 3.0	Time: $O(E \log(E) + Ek_n)$ where, k_n = degree of a node. Space: $O(E + V \log(E))$ If the output is stored 2-columns at a time directly in secondary storage, it reduces to $O(E + V)$.
ForceAtlas2 [61], [62]	- Energy Force-directed layout algorithm - Nodes repel each other, edges attract nodes - Utilized in the GEPHI tool. - Produces good quality visualization	Time: N/A Space: N/A
Kamada-Kawai with multiple node selection (KK-MS) [63], [64]	- Force-directed layout algorithm - Modifies by updating some nodes at every iteration in the Kamada-Kawai algorithm. - Evaluated based on accuracy. - Utilizes the concept of Neighborhood matching	Time: $O(n(V + v^2))$ where, n = number of iterations Space: N/A
Kamada-Kawai with multiple node selection and decaying stiffness (KK-MS-DS) [63]	- The idea is to reduce the amount of time taken for the KK algorithm - Unlike the KK algorithm, stiffness is no longer a constant - Has higher sensitivity and specificity than KK-MS	Time: N/A Space: N/A

- Topological Structure Recognition
- Structural Hole Analysis
- Graph Colouring
- Graph Partitioning
- Dynamic Network Analysis

For example, influence propagation involves detecting the direction of information flow within nodes whose opinions are most received throughout the social network [65]. Companies like YouTube have used this task to identify and foster the *influencer trend*. This way, companies know which *Youtubers* to select to advertise their products for maximum reach. Depending on the approach a practitioner takes to influence propagation, similarity graph algorithms could be utilized alongside clustering algorithms to determine the flow of information. Similarity algorithms (e.g., Jaccard Coefficient) will help identify nodes with similar features, while clustering algorithms will be used to determine the sphere of influence. Hence, we cannot say that a strict

classification or taxonomy of these algorithms could be determined from our survey. Consequently, a practitioner seeking to employ the use of these algorithms must first discover:

- 1) What output is sought?
- 2) What application area they are involved in?
- 3) What computational resources do they have available?

In general, we recommend the steps outlined in Figure 8 for Social Network Analysis practitioners:

Once a practitioner answers these basic questions, this survey will help them determine what tasks they can carry out, some of the more recent algorithms they could use, and in what combination to achieve their desired results, given their resource constraints.

Tables 1 - 6 summarize some of the real-life applications of the graph algorithms in SNA. They also outline the complexities stated in the reference paper and the resultant operational aspects recommended for the algorithms by

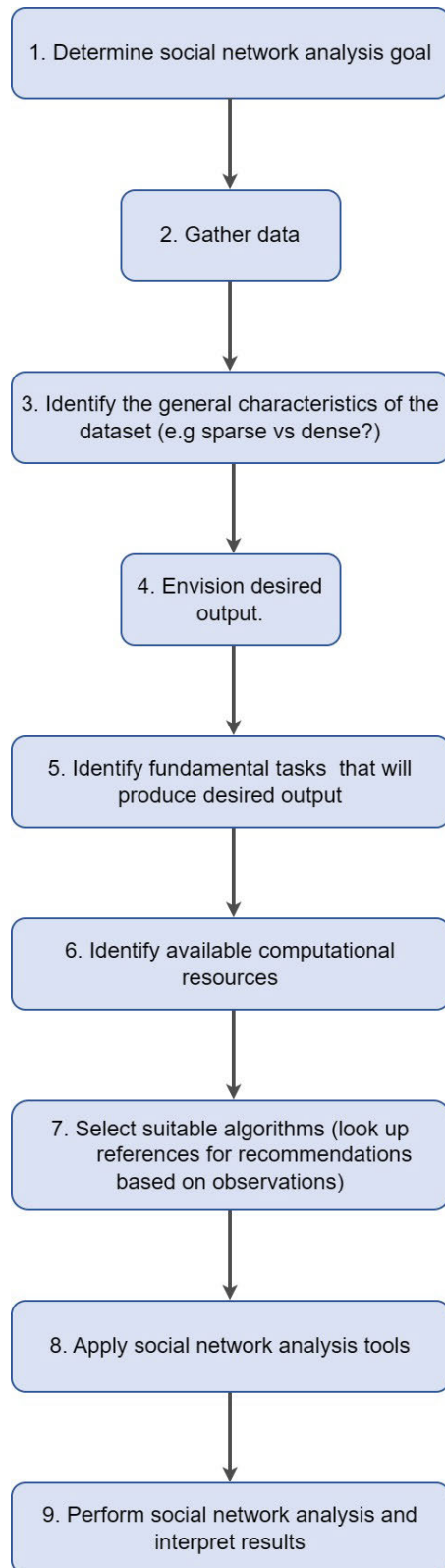


FIGURE 8. Recommended steps for Social Network Analysis (SNA).

the authors. These complexities, operational aspects, and references have provided easy guidelines.

VII. CASE STUDY: FRAUD DETECTION DATASET

In this case study, we demonstrate the procedure we have recommended for SNA and the usability of this survey by identifying patterns within a fraud detection dataset publicly hosted on Kaggle [66]. Fraudulent financial transactions are a common issue for all. From financial institutions to governmental bodies, financial fraud constitutes a substantial economic loss, costing hundreds of billions in dollars [67].

A. STEP 1: DETERMINE SOCIAL NETWORK ANALYSIS GOAL

Our hypothetical goal for this SNA endeavor is to identify possible patterns of organized fraudulent transactions or central fraud actors and the like. Furthermore, we intend to identify common patterns or characteristics that could signify a fraudulent transaction. Our SNA goal is comprised of the following specific objectives, leveraging graph algorithms:

- Identify recurring subgraphs.
- Identify centralized nodes.
- Extract recurring structural graph features.

B. STEP 2: GATHER DATA

The fraud detection dataset we have chosen contains about 50,000 unique financial transactions, marked as either fraudulent or lawful, and derived from real-world scenarios [66]. Our data comprises the following features:

- 1) Transaction ID: A unique alphanumeric identifier for the transaction.
- 2) User ID: A unique numeric identifier for the user.
- 3) Transaction Amount: The amount involved in the transaction stated in floating-point datatype for decimal precision. No specific currency is stated for the amounts directly in the dataset, but we can safely assume dollars based on the transaction locations listed.
- 4) Transaction Type: This refers to the type of transaction, ranging from ATM withdrawals, bill payments, bank transfers, online purchases, and POS payments.
- 5) Time of Transaction: This is the time at which the transaction was conducted, recorded in the 24-hr time format.
- 6) Device Used: Type of device used to conduct the transaction - mobile, desktop, etc.
- 7) Location: Originating location for the transaction in the range of states in the US.
- 8) Previous Fraudulent Transactions: The author of the data did not explicitly explain the indication of this column. However, from observation, it appears to be a value ranging between 0 and 4, most likely indicating the level of fraud risk, given the specifics of the transaction. 0 represents no risk, and 4 indicates the highest risk approximation. Furthermore, given the possibility that this data is synthetic, it could also be noise.
- 9) Account Age: This indicates the age of the user account. The unit for this measurement is not explicitly

stated by the author, but its values range between 1 and 119. It is most likely measured in days, as new accounts under the 3-month range are most often vetted for fraudulent transactions.

- 10) Number of Transactions in the last 24 hours: Lists the number of transactions done by the user in the last 24 hours.
- 11) Payment Method: Payment method involved in the transaction - credit card, debit card, and Unified Payments Interface (UPI) - digital payments specific to India. Given that the transaction locations listed are all in the United States, the fact that UPI is listed could indicate that the dataset is synthetic.
- 12) Fraudulent: This column marks the transaction as fraudulent or not. The values in this column are binary classification - 0 (not fraudulent) and 1 (fraudulent).

The procedures, techniques, and graph algorithms applied to this dataset are scalable to large datasets encompassing even millions of units of data.

C. STEP 3: IDENTIFY GENERAL CHARACTERISTICS OF THE DATASET

We previously identified several data types within the dataset, ranging from numeric (user ID, transaction amount, etc.) to alphanumeric (transaction ID) and categorical text data (device used, payment method, etc.). Our dataset comprises 51000 records of transactions with transaction amounts ranging from 5 to 50000. The dataset is dense, as validated by the few null values. The transaction ID and user ID columns serve as unique identifiers for the dataset. Aside from the *fraudulent* and unique ID columns, all others are shared entities and could potentially provide edges/relationships in our graph.

From these characteristics, we can see that graphical analysis is feasible with this dataset and will provide usable insights that lead to the fulfillment of our SNA goal. The density of the dataset indicates that working through the dataset might be computationally expensive. We must account for this while choosing our graph algorithms. We could also cater to this by implementing optimization techniques, but we will tackle recommendations for this scenario in further steps.

Our dataset has some time-based data. This opens up opportunities for possible temporal graphical analysis to identify the effects time has on the probability of fraudulent transactions. These are all observable characteristics from the dataset. Observing these characteristics before further analysis will guide our selection of graph representations and ensure we use relevant graph algorithms. Armed with the characteristics of the dataset, we can avoid garbage-in, garbage-out results from our analysis.

D. STEP 4: ENVISION DESIRED OUTPUT

Now that we have gleaned some general characteristics from the dataset, we can envisage the graphical outputs that could fulfill the stated SNA goals.

For the goal of identifying recurring subgraphs within the dataset, the graph representations for any potential recurring subgraphs within the fraud dataset could include some repeated transaction flows, maybe with similar attributes (location, time of transactions), which will help identify outliers or suspicious recurrent trends like specific transaction amounts at particular times of the day or only on specific devices.

To achieve the goal of identifying centralized nodes, the expected output should highlight nodes that are highly connected within the fraudulent or non-fraudulent transaction subnetwork, or across the entire dataset. Depending on the node feature category of the highlighted nodes, we could then make meaningful correlation assumptions from our dataset.

The goal of identifying recurring structural features should reveal patterns in the topology of our dataset. These patterns could include prominent motifs like open or closed triads, feed-forward loops, etc. They could also be community clusters, and we could infer attribute-correlated structures between our dataset features or determine bridges, hubs, or brokers in the dataset.

In summary, envisioning our results from the proposed analysis of the fraud dataset will guide us in the quick and specific selection of appropriate algorithms.

E. STEP 5: IDENTIFY FUNDAMENTAL TASKS THAT WILL PRODUCE DESIRED OUTPUT

Given the goals of the fraud dataset analysis (see VII-A) and the desired results (see VII-D), we can determine which of our listed fundamental SNA tasks will yield the intended outcome. Depending on the computational resources available, we may need to employ graph sampling to study representative subsets rather than the 50,000 data points. To achieve our goals, we will also leverage centrality, community detection, and graph visualization.

F. STEP 6: IDENTIFY AVAILABLE COMPUTATIONAL RESOURCES

The hardware resources we have available for the analysis include 16 GB of RAM, a limited Intel Xe GPU, and a Core i7 CPU. This implies that the memory may not support computationally expensive adjacency matrices; only sparse adjacency lists are allowed. Also, parallel processing capabilities are limited. A sufficient amount of storage to the tune of 500 GB is available. The software resources we have available for the analysis include a Windows 10 OS with no access to distributed systems. We have support for most programming languages and supporting libraries.

G. STEP 7: SELECT SUITABLE ALGORITHMS

Based on the information gleaned from previous steps, we can make informed graph algorithm choices. We have identified that, to achieve our goals, we will implement centrality, similarity, graph sampling, and graph visualization. For the centrality task, the degree centrality algorithm and PageRank algorithms are applicable and fit our resource availability.

Also, for community detection tasks, we can use the Louvain and Clique-Louvain algorithms to address time-complexity issues. Random Walk Sampling and Snowball Sampling algorithms will cover the graph sampling tasks. Informed by our references, these algorithms are best suited to our goals, envisioned output, and available resources.

H. STEP 8: APPLY A SOCIAL NETWORK ANALYSIS TOOL

There is a diverse array of tools available for SNA. Desktop or web-based GUI SNA tools such as Gephi or NodeXL are intuitive, enabling a shallow learning curve and quick results. Programmatic tools employ popular libraries such as NetworkX and igraph and allow for ease of customization and reproducibility. For our case study, we will be using Python-based programmatic tools and the NetworkX and igraph libraries to perform analysis on the fraud dataset. They best suit our available computational resources.

I. STEP 9: PERFORM SOCIAL NETWORK ANALYSIS AND INTERPRET RESULTS

Our dataset is most likely synthetic. Preliminary analysis revealed that no dataset feature displayed a distinguishing relationship with our target variable, *fraudulent*. Node-to-node relationships between the target variable and other features (account age, time of transaction, transaction amount, transaction type, device used, etc.) are evenly weighted, with no standout patterns. Regardless, our case study successfully demonstrates the practical step-by-step procedure to SNA, as earlier hypothesized.

Limitations of Graph Algorithms for Social Network Analysis: Graph algorithms utilize structural and topological information. Combining their structural analysis advantage with content analysis (for example, user profile information, nationality, etc.) will yield a more holistic view of the network.

Furthermore, the use of machine learning models for SNA must be addressed. Although this paper did not focus on those aspects, it offers another methodology for SNA. [68]. Deep learning-based approaches provide further insights when used alongside the mentioned algorithms. Abbas, Ash Mohammed [69] conducts a detailed assessment of the state-of-the-art in deep learning techniques for SNA and their application areas.

VIII. CONCLUSION

Social Network Analysis is an interdisciplinary field that deals with exploring and understanding network structures. Graphs have historically been used to represent social networks, so the field depends heavily on graph algorithms. SNA tasks include community detection, similarity, clustering, centrality, graph sampling, and link prediction, and several graph algorithms are used in these tasks. Each algorithm has unique application areas and time and resource constraints that contribute to SNA. This survey provides a comprehensive document for both practitioners and researchers to start

performing SNA. It covers the required resource considerations and a basic understanding of the graph algorithms.

We have provided an expository survey of the role of graph algorithms in SNA with an aim to synthesize the multitude of available information on SNA and its corresponding graph algorithms into a single document that can serve as a starter for any party with an interest in SNA, expert or not. This paper focuses on the more recent algorithms within the field. Given the multitude of graph algorithms within the SNA field, they cannot be completely captured within one survey. But this work provides a good foundation for the subject.

The SNA tasks encountered in this survey included centrality, graph sampling, link prediction, community detection/clustering, similarity, and graph visualization. In conclusion, these graph algorithms are crucial to social network analysis.

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